Collision-Based Export Formula Report

Generated from stress_collision_based.py

Formulas implemented in `stress_collision_based.py`

This document summarizes the exact formulas currently implemented in `stress_collision_based.py` to produce exported CSV data.

Output files

Type	File name pattern	Content
Summary CSV	`<dem>_collision_over_time.csv`	One row per processed timestep.
Extracted CSV	`<dem>_collision_extracted_at_<time>.csv`	Per-collision values at the selected extract time.

Notation

Symbol	Meaning
k	Current timestep index.
t_k	Current sample time returned by EDEM for timestep k.
Δt_k	Interval length from the previous sample to the current sample.
C_k	Set of collisions present in the current surfSurf sample.
$N_k = \lvert C_k \rvert$	Number of sampled collisions at timestep k.
N_k^{new}	Number of sampled collisions whose start time lies inside the current interval.
s_i, e_i	Start time and end time of collision i reported by EDEM.
$\tau_i^{overlap}$	Overlap duration of collision i with the interval $[t_{k-1}, t_k]$.
d_i	Full contact duration of collision i.
$I[\cdot]$	Indicator function equal to 1 when the stated condition is true and 0 otherwise.
$m_{1,i}, m_{2,i}$	Masses of the two particles in collision i.
$r_{1,i}, r_{2,i}$	Radii of the two particles in collision i.
$\mathbf{F}_{n,i}, \mathbf{F}_{s,i}$	Current total normal and total tangential force vectors from EDEM.
$F_{n,i}, F_{s,i}, F_i$	Magnitudes of normal, tangential, and total collision force used in the code.
F_i^{max}	Maximum total collision-force magnitude formed from stored peak-force vectors.

Symbol	Meaning
$E_{n,i}, E_{s,i}, E_i$	Normal, shear, and total collision energies used in the code.
$m_{eff,i}$	Reduced mass of collision i.
$J_{n,i}, J_{s,i}, J_i$	Normal, shear, and total impulse attributed to the current interval.
A_i	Collision area proxy used by the exported stress calculation.
$\sigma_{n,i}, \sigma_{s,i}, \sigma_i$	Exported normal, shear, and total stress-like values per collision.
$P_{n,i}, P_{s,i}, P_i$	Interval-average power values per collision.
$n_{h,k}$	Number of current sampled collisions touching host particle h at timestep k.
$\boldsymbol{\omega}_h$	Angular-velocity vector of host particle h.
$d^{rot}_{h,k}, D^{rot}_{h,k}$	Current and cumulative rotational travel distance for host particle h.
$\xi_{h,k}$	Binary indicator showing whether host particle h is contacting at timestep k.
$T^{contact}_{h,k}$	Cumulative time for which host particle h is marked as contacting.
H_k	Host particles selected by the configured host material.

How to read the symbols

Time and interval symbols k is a sample index and t_k is the physical simulation time at that sample. Δt_k is the reporting interval width, so it is the denominator for rate-type quantities such as collision frequency and power. Each collision has a start time s_i and end time e_i, but only the overlap part $\tau_i^{overlap}$ is assigned to the current interval $[t_{k-1}, t_k]$.

Mechanical symbols $m_{1,i}, m_{2,i}$ and $r_{1,i}, r_{2,i}$ are the masses and radii of the two particles forming collision i. Bold symbols $\mathbf{F}_{n,i}$ and $\mathbf{F}_{s,i}$ are vectors from EDEM, whereas $F_{n,i}$, $F_{s,i}$, and F_i are magnitudes used by the export code. E terms denote energies, $m_{eff,i}$ is the reduced mass that normalizes those energies into specific intensity, and J terms are impulses obtained by integrating force over time.

Stress and power symbols A_i is a geometric area proxy built from particle radii, not a true Hertzian contact patch. The exported stress-like values $\sigma_{n,i}$, $\sigma_{s,i}$, and σ_i come from impulse divided by overlap time and by this area proxy. $P_{n,i}$, $P_{s,i}$, and P_i are interval-average powers computed by dividing collision energy by Δt_k.

Host-particle rotation symbols For a host particle h, $\boldsymbol{\omega}_h$ is angular velocity, $n_{h,k}$ is the number of active sampled contacts touching that host particle, and $\xi_{h,k} = I[n_{h,k} > 0]$ converts that contact state into a binary gate. That gate decides whether the current rotational travel distance $d^{rot}_{h,k}$ contributes to the cumulative rotational distance $D^{rot}_{h,k}$ and cumulative contact time $T^{contact}_{h,k}$.

Core equations

C1. Interval length

$$\Delta t_k = \max(0, t_k - t_{k-1})$$

For the first processed timestep, the implementation uses 0 as the previous time if no earlier sample is available.

C2. Overlap duration inside the current interval

$$\tau_i^{overlap} = \max(0, \min(e_i, t_k) - \max(s_i, t_{k-1}))$$

This is the contact time attributed to the current interval for collision i.

C3. Full contact duration of one collision

$$d_i = \max(0, e_i - s_i)$$

Used for the summary field `Ave_contact_duration_per_collision`.

C4. Collision frequency based on new collisions started in the interval

$$N_k^{new} = \begin{cases} \sum_{i \in C_k} I[t_{k-1} \le s_i \le t_k], & t_{k-1} \le 0, \\ \sum_{i \in C_k} I[t_{k-1} < s_i \le t_k], & t_{k-1} > 0, \end{cases}$$
$$f_k^{coll} = \frac{N_k^{new}}{\Delta t_k}$$

N_k^{new} is computed exactly as in `count_started_collisions` in the code: the first interval includes the lower bound so collisions starting exactly at time 0 are counted once, while later intervals use an open lower bound to avoid double counting collisions that start exactly on a timestep boundary.

C5. Current force magnitudes

$$F_{n,i} = \|\mathbf{F}_{n,i}\|,$$
$$F_{s,i} = \|\mathbf{F}_{s,i}\|,$$
$$F_i = \|\mathbf{F}_{n,i} + \mathbf{F}_{s,i}\|$$

The total collision force magnitude is the norm of the vector sum, not the sum of norms.

C6. Maximum-force magnitude used for impact statistics

$$F_i^{max} = \|\mathbf{F}_{n,i}^{max} + \mathbf{F}_{s,i}^{max}\|$$

Used for `Ave_Fcol_Max_per_collision`, `Max_Fcol_Max_per_collision`, and `high_impact_event_fraction`.

C7. Reduced mass

$$m_{eff,i} = \frac{m_{1,i} m_{2,i}}{m_{1,i} + m_{2,i}}$$

If the denominator is zero, the code treats the reduced mass as invalid and writes zero for specific-intensity terms.

C8. Collision specific-intensity values

$$SI_{n,i} = \frac{E_{n,i}}{m_{eff,i}},$$
$$SI_{s,i} = \frac{E_{s,i}}{m_{eff,i}},$$
$$SI_i = \frac{E_i}{m_{eff,i}}$$

Here $E_i = E_{n,i} + E_{s,i}$. The current code sets dissipation energy to zero.

C9. Collision area proxy used in stress calculations

$$A_i = \pi \left(\frac{r_{1,i} + r_{2,i}}{2} \right)^2$$

This is a geometric proxy area from the current implementation, not a Hertz contact patch.

C10. Trapezoidal impulse for matched sampled collisions

$$\begin{aligned} J_{n,i} &= \frac{1}{2} \left(F_{n,i}^{prev} + F_{n,i}^{curr} \right) \tau_i^{overlap}, \\ J_{s,i} &= \frac{1}{2} \left(F_{s,i}^{prev} + F_{s,i}^{curr} \right) \tau_i^{overlap}, \\ J_i &= \frac{1}{2} \left(F_i^{prev} + F_i^{curr} \right) \tau_i^{overlap} \end{aligned}$$

The code matches current and previous collisions by unordered particle pair and, when needed, nearest contact position.

C11. Collision stress values from impulse, overlap time, and area

$$\begin{aligned} \sigma_{n,i} &= \frac{J_{n,i}}{\tau_i^{overlap} A_i}, \\ \sigma_{s,i} &= \frac{J_{s,i}}{\tau_i^{overlap} A_i}, \\ \sigma_i &= \frac{J_i}{\tau_i^{overlap} A_i} \end{aligned}$$

If overlap time or area is non-positive, the code writes zero.

C12. Collision power values

$$\begin{aligned} P_{n,i} &= \frac{E_{n,i}}{\Delta t_k}, \\ P_{s,i} &= \frac{E_{s,i}}{\Delta t_k}, \\ P_i &= \frac{E_i}{\Delta t_k} \end{aligned}$$

Power uses the current interval length in the denominator.

C13. Aggregate specific energy by collision

$$\begin{aligned} SE_{n,k}^{coll} &= \frac{\sum_{i \in C_k} E_{n,i}}{\sum_{i \in C_k} m_{eff,i}}, \\ SE_{s,k}^{coll} &= \frac{\sum_{i \in C_k} E_{s,i}}{\sum_{i \in C_k} m_{eff,i}}, \\ SE_k^{coll} &= \frac{\sum_{i \in C_k} E_i}{\sum_{i \in C_k} m_{eff,i}} \end{aligned}$$

The implementation aggregates energy first and then divides by the sum of valid reduced masses.

C14. Particle split used for particle SI values still exported in the collision summary CSV

$$E_{n,p} = \frac{1}{2}\sum_{i\sim p} E_{n,i},$$
$$E_{s,p} = \frac{1}{2}\sum_{i\sim p} E_{s,i},$$
$$SI_{n,p} = \frac{E_{n,p}}{m_p},$$
$$SI_{s,p} = \frac{E_{s,p}}{m_p}$$

These particle-based SI values appear in `collision_over_time.csv` because the shared timestep core computes both collision and particle summaries.

C15. Host-particle rotational metrics

$$d_{h,k}^{rot} = r_h \|\boldsymbol{\omega}_h\| \Delta t_k,$$
$$\xi_{h,k} = I[n_{h,k} > 0],$$
$$D_{h,k}^{rot} = D_{h,k-1}^{rot} + \xi_{h,k} d_{h,k}^{rot},$$
$$T_{h,k}^{contact} = T_{h,k-1}^{contact} + \xi_{h,k}\Delta t_k$$

The contact indicator uses the current sampled active collision count for each host particle.

Summary CSV fields

Field	Calculation / reference	Meaning
`time`	Current sample time t_k.	EDEM time value for the processed timestep.
`number_of_collisions`	$N_k = \|C_k\|$.	Number of sampled collisions in the current surfSurf snapshot.
`Ave_contact_duration_per_collision`	Mean of d_i over current sampled collisions. Eq. C3.	Average full contact duration.
`total_collision_time`	Sum of $\tau_i^{overlap}$ over sampled collisions. Eq. C2.	Total overlap time attributed to the current interval.
`average_collision_time`	Mean of $\tau_i^{overlap}$ over sampled collisions. Eq. C2.	Average overlap time per sampled collision in the current interval.
`collision_frequency`	$N_k^{new}/\Delta t_k$. Eq. C4.	Frequency of newly started sampled collisions in the interval.
`Ave_TotalColForce_inSys_byCollision`	Mean of F_i over sampled collisions. Eq. C5.	Average current total collision force magnitude.
`Ave_Fcol_Max_per_collision`	Mean of F_i^{max}. Eq. C6.	Average maximum force magnitude.
`Max_Fcol_Max_per_collision`	Max of F_i^{max}. Eq. C6.	Largest maximum force magnitude in the sampled collision set.

Field	Calculation / reference	Meaning
`high_impact_event_fraction`	Count(F_i^{max} > threshold) / N_k. Eq. C6.	Fraction of sampled collisions above the configured impact threshold.
`Ave_SI_normal_per_collision`	Mean of $SI_{n,i}$ over valid reduced masses. Eq. C8.	Average collision normal specific intensity.
`Ave_SI_shear_per_collision`	Mean of $SI_{s,i}$ over valid reduced masses. Eq. C8.	Average collision shear specific intensity.
`Ave_SI_total_per_collision`	Mean of SI_i over valid reduced masses. Eq. C8.	Average collision total specific intensity.
`Total_normal_specific_energy_by_collision`	Aggregate collision normal specific energy. Eq. C13.	Energy summed over collisions, then divided by summed reduced mass.
`Total_shear_specific_energy_by_collision`	Aggregate collision shear specific energy. Eq. C13.	Energy summed over collisions, then divided by summed reduced mass.
`Total_specific_energy_by_collision`	Aggregate collision total specific energy. Eq. C13.	Energy summed over collisions, then divided by summed reduced mass.
`Ave_SI_normal_per_particle`	Mean over particles of $SI_{n,p}$ from the shared core. Eq. C14.	Particle-based normal SI still written into the collision summary CSV.
`Ave_SI_shear_per_particle`	Mean over particles of $SI_{s,p}$ from the shared core. Eq. C14.	Particle-based shear SI still written into the collision summary CSV.
`Ave_SI_total_per_particle`	Mean over particles of SI_p from the shared core. Eq. C14.	Particle-based total SI still written into the collision summary CSV.
`Ave_normal_stress_value_per_collision`	Mean of $\sigma_{n,i}$ over sampled collisions. Eq. C11.	Average normal stress proxy per collision.
`Ave_shear_stress_value_per_collision`	Mean of $\sigma_{s,i}$ over sampled collisions. Eq. C11.	Average shear stress proxy per collision.
`Ave_stress_value_per_collision`	Mean of σ_i over sampled collisions. Eq. C11.	Average total stress proxy per collision.
`Ave_normal_power_per_collision`	Mean of $P_{n,i}$ over sampled collisions. Eq. C12.	Average normal power per collision.
`Ave_shear_power_per_collision`	Mean of $P_{s,i}$ over sampled collisions. Eq. C12.	Average shear power per collision.
`Ave_power_per_collision`	Mean of P_i over sampled collisions. Eq. C12.	Average total power per collision.
`Total_normal_power_by_collision`	Sum of $P_{n,i}$ over sampled collisions. Eq. C12.	Total normal power across the current sampled collision set.

Field	Calculation / reference	Meaning
`Total_shear_power_by_collision`	Sum of $P_{s,i}$ over sampled collisions. Eq. C12.	Total shear power across the current sampled collision set.
`Total_power_by_collision`	Sum of P_i over sampled collisions. Eq. C12.	Total power across the current sampled collision set.
`Ave_rotational_distance_per_host_particle`	Mean of $d^{rot}_{h,k}$ over all host particles sampled at k. Eq. C15.	Average host rotational distance in the current interval.
`Ave_rotational_distance_total_per_host_particle`	Mean of $D^{rot}_{h,k}$ over host particles with $\xi_{h,k} = 1$. Eq. C15.	Average cumulative host rotational distance among currently contacting host particles.
`Ave_cumulative_contact_time_per_host_particle`	Mean of $T^{contact}_{h,k}$ over host particles with $\xi_{h,k} = 1$. Eq. C15.	Average cumulative host contact time among currently contacting host particles.

Extracted CSV fields

Field	Calculation / reference	Meaning
`time`	Scalar t_k for the extract timestep.	Written once at the top of the extracted block by the CSV expansion helper.
`collision_id`	Row index $0, 1, \ldots, N_k - 1$.	Local collision row number in the current extract snapshot, not a persistent collision ID.
`first_particle_id`	Raw first collision particle ID from EDEM.	First particle in the sampled collision pair.
`second_particle_id`	Raw second collision particle ID from EDEM.	Second particle in the sampled collision pair.
`specific_energy_normal_per_collision`	$SI_{n,i}$ when reduced mass is valid, else 0. Eq. C8.	Collision normal specific intensity.
`specific_energy_shear_per_collision`	$SI_{s,i}$ when reduced mass is valid, else 0. Eq. C8.	Collision shear specific intensity.
`specific_energy_total_per_collision`	SI_i when reduced mass is valid, else 0. Eq. C8.	Collision total specific intensity.
`collision_area`	A_i from Eq. C9.	Collision area proxy used by the stress calculation.
`normal_force_magnitude_per_collision`	$F_{n,i}$ from Eq. C5.	Current normal-force magnitude.
`shear_force_magnitude_per_collision`	$F_{s,i}$ from Eq. C5.	Current tangential-force magnitude.
`total_force_magnitude_per_collision`	F_i from Eq. C5.	Current total collision-force magnitude.

Field	Calculation / reference	Meaning
`normal_stress_value_per_collision`	$\sigma_{n,i}$ from Eq. C11.	Normal stress proxy.
`shear_stress_value_per_collision`	$\sigma_{s,i}$ from Eq. C11.	Shear stress proxy.
`stress_value_per_collision`	σ_i from Eq. C11.	Total stress proxy.
`normal_power_per_collision`	$P_{n,i}$ from Eq. C12.	Normal power.
`shear_power_per_collision`	$P_{s,i}$ from Eq. C12.	Shear power.
`power_per_collision`	P_i from Eq. C12.	Total power.

Implementation notes

- All displayed formulas reflect the current implementation in `stress_collision_based.py`.
- Collision continuity for the trapezoidal rule is matched by unordered particle pair and nearest contact position when multiple contacts share the same pair.
- The collision summary and collision extracted file do not add a separate previous-only tail term. Tail handling is only added at particle level inside the shared timestep core.
- If a denominator is zero, if no valid term exists, or if a feature flag is disabled in `Stress_settings.txt`, the corresponding exported values remain 0.
- The CSV expansion helper writes scalar fields such as time only once at the top of an extracted block, leaving the remaining rows blank in that scalar column.